(c) There is no hop-by-hop segmentation procedure in IPv6. Therefore there is no fragment offset, flags and identification fields in IPv6. There is also no need for header length field.

(d) Type of service (TOS) field of IPv4 is removed in IPv6. This removal is also based on experience, as TOS field is hardly used in IPv4.

- IPv6 defines six extension headers: hop-by-hop options, routing, fragment, encrypted security payload, authentication and destination options.
- IPv6 has different options, security and easy transition facility.

Technically, IPv6 can be compared with IPv4 in terms of two important parameters, namely, address space and IP header simplification. A few people projected the death of the Internet due to increased traffic, yet with IPv6, Internet access is guaranteed, with exponential increase in Internet users over time with the provision of abundant address space of 2^{128}.

With a simplified header description, IPv6 is prone to provide multimedia (voice, video and data) services. It is therefore the future replacement of IPv4.

It was felt that a new version after IPv4 should take care of more addresses, reduced overhead, better routing, support for address renumbering, improved header processing, reasonable security, and support for multimedia and mobility. IPv6 has two new fields: flow label and priority. These are included to facilitate the handling of real-time services like voice, video and quality multimedia services.

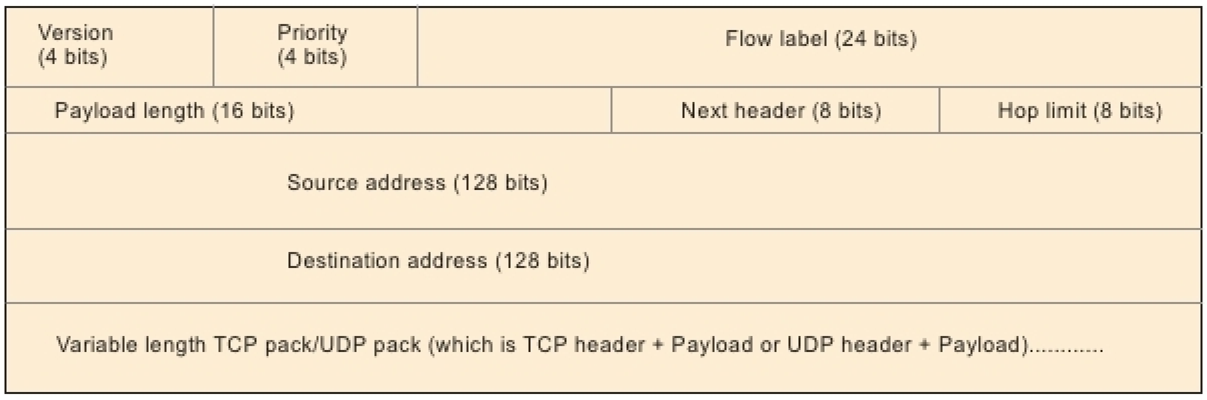

Figure 1: IPv6 packet format

About IPv6

Salient features of IPv6, also shown in Figure 1, may be summarised as:

- IPv6 has fixed header bytes like that in an ATM cell, which is of fixed 53 bytes. What is the advantage of having a fixed size? Let us consider the postal system. If envelopes are of variable sizes and weights, processing gets delayed, as the process includes measuring weight, verifying postal stamps required according to weight and so on. If there were fixed-sized and weighted envelopes, postal processing would take much less time and effort. Similarly, with a fixed header, node processing delay is reduced. This makes IPv6 more suitable for real-time services like voice, video and multimedia.
- A 128-bit address space in IPv6, instead of a 32-bit address space in IPv4, is to ensure that there will be no exhaustion of Internet address space. It is expected that the Internet under IPv6 supports 10^{15} (quadrillion) hosts and 10^{12} (trillion) networks. The Internet under IPv4 can support maximum 2^{32} hosts. IPv6 address space is about 2^{96} times more than that of IPv4. This is how it can ensure that exhaustion of address space is almost unlikely with IPv6.
- Besides unicast and multicast, IPv6 has the provision for anycast addressing that allows a packet addressed to an anycast address to be delivered to any one of the group of hosts.
- IPv6 has the provision for a single address being associated with multiple interfaces.
- Address auto configuration and classless inter-domain routing (CIDR) addressing are provided in IPv6.
- Provision of extension header in IPv6 meets the requirements of such things as checksum, security options, etc.
- Flow-level and priority headers are used to support real-time services. By assigning higher priority to real-time packets, necessity of time sensitiveness is restored. Data packets and, for that purpose, time-insensitive packets are assigned low priority and serviced by the best effort approach.
- IPv6 ensures support for real-time services.
- IPv6 provides security support that could eventually be seen as its biggest advantage. Today, billion-dollar businesses are carried out over the Internet. To keep businesses secure, a public crypto system has emerged as one of the most important tools. IPv6 with its ancillary security protocol has provided a better communication tool for transacting business activities over the Internet.
- IPv6 has the provision for enhanced routing capability, including support for mobile hosts.

Functions of IPv6 headers bytes are given below.

- Version field (4 bits) contains the version number of Internet Protocol. As of now, there are only two versions: 4 and 6. For version 6, this field is six, which makes four bits of this field as 0110.
- Priority (4 bits) field is used to indicate the priority of the datagram. Four bits in the field define 2^4(=16) priority levels. The first eight priority levels (from 0000 to 0111) are assigned for the services requiring congestion control. If congestion occurs, traffic is backed off.

These are suitable for conventional data and non-real-time services. Different priority levels under the first eight levels are used as 0 for no priority, 1 for background traffic like net-news, 2 for unattended transfer like e-mails, 3 remains reserved, 4 for attended bulk transfer like File Transfer Protocol (FTP), 5 remains reserved, 6 for interactive traffic such as Telnet, X-windows and so on, and 7 for control traffic such as Simple Network Management Protocol (SNMP) and routing protocols.